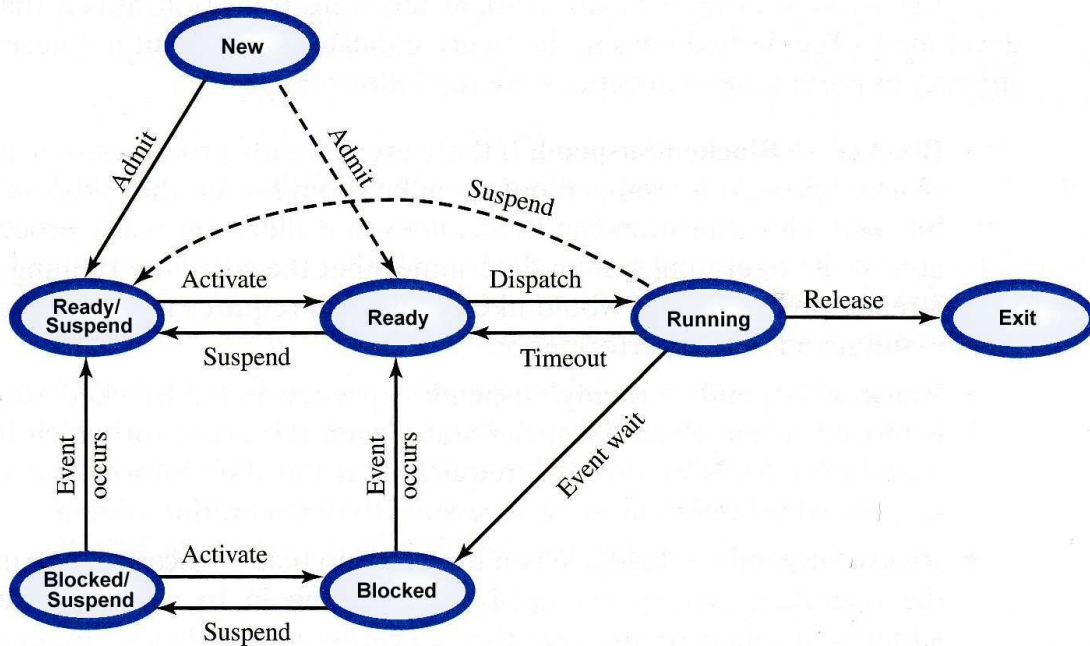


3 – Stallings 3.4

Consider the state transition diagram in the figure. Suppose that it is time for the operating system to dispatch a process and that there are processes in both the Ready state and the Ready/Suspend state, and that at least one process in the Ready/Suspend state has higher scheduling priority than any of the processes in the Ready state. Two extreme policies are:

1. Always dispatch from a process in the Ready state, to minimize swapping.
2. Always give preference to the highest-priority process, even though that may mean swapping when swapping is not necessary.

Suggest an intermediate policy that tries to balance the concerns of priority and performance.



- Ready: The process is in main memory and available for execution.
- Blocked: The process is in main memory and awaiting an event.
- Blocked/Suspend: The process is in secondary memory and awaiting an event.
- Ready/Suspend: The process is in secondary memory but is available for execution as soon as it is loaded into main memory.

4 – Stallings 3.9

In a number of early computers, an interrupt caused the register values to be stored in fixed locations associated with the given interrupt signal. Under what circumstances is this a practical technique? Explain why it is inconvenient in general.